

31 Linearly independent reactions

Why Are We Learning This?

In the previous lectures, we learned how to represent reactions using stoichiometry matrices and how reaction mechanisms are built from elementary reaction steps. However, not every reaction in a mechanism provides new information. Some reactions can be generated by combining other reactions already present in the mechanism. This leads to the concept of **linear independence**.

Knowing how many linearly independent reactions exist is important because it tells us:

- How many truly independent pathways exist in a mechanism.
- Whether some reactions are redundant.
- Whether the mechanism has been specified efficiently.
- How to simplify large reaction networks.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



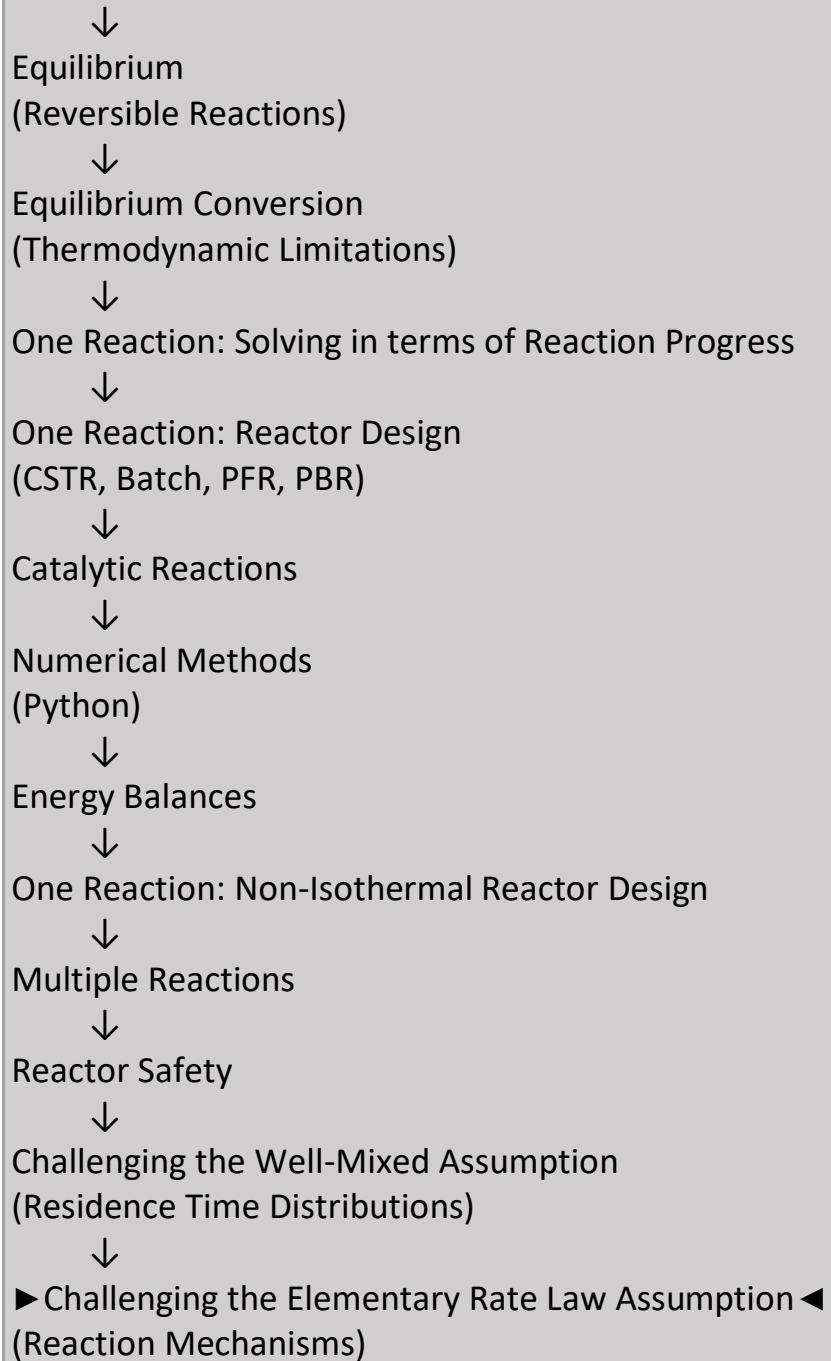
Reaction Progress
(Conversion)



Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



Rate Laws
(Kinetics)



Learning Objectives

By the end of this lecture, you should be able to:

- Define linearly independent and linearly dependent reactions.

- Explain the relationship between reaction pathways and linear independence.
- Determine the number of independent reactions in a reaction network.
- Construct a stoichiometry matrix for a reaction set.
- Use matrix rank to determine the number of independent reactions.
- Identify reactions that can or cannot be eliminated from a proposed reaction set.
- Understand why linear dependence frequently occurs in real reaction mechanisms.

Big Idea

Reaction Mechanism



Stoichiometry Matrix



Matrix Rank

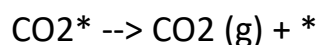
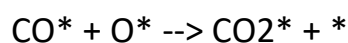
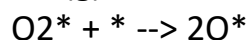
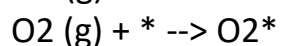
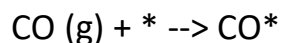


Number of Independent Reactions

Linear independence tells us how much unique stoichiometric information is contained within a reaction mechanism.

I found [this video](#) helpful when writing this lecture.

The following is an example of a linearly independent set of reactions :



This mechanism represents a "straight path" from reactants to products, i.e., that there is only one route from reactants to products.

What Does "Linearly Independent" Mean?

A reaction is linearly independent if it cannot be generated by adding, subtracting, or multiplying other reactions in the set.

Physical Interpretation

Each independent reaction provides: New Chemical Information that is not contained in any other reaction.

Example: Reaction 1, Reaction 2, Reaction 3 are linearly independent if none of them can be reproduced using combinations of the others.

Every Reaction Adds Something New to the mechanism.

Why Is This Mechanism Linearly Independent?

The mechanism represents a single pathway from reactants to products. Each reaction step performs a unique function: Adsorption, Activation, Surface Reaction, and Desorption. None of the steps can be constructed by combining other steps in the mechanism.

Physical Interpretation

Step 1

↓

Step 2

↓

Step 3

↓

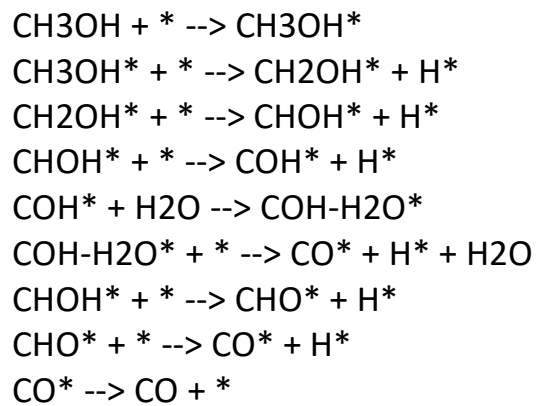
Step 4

↓

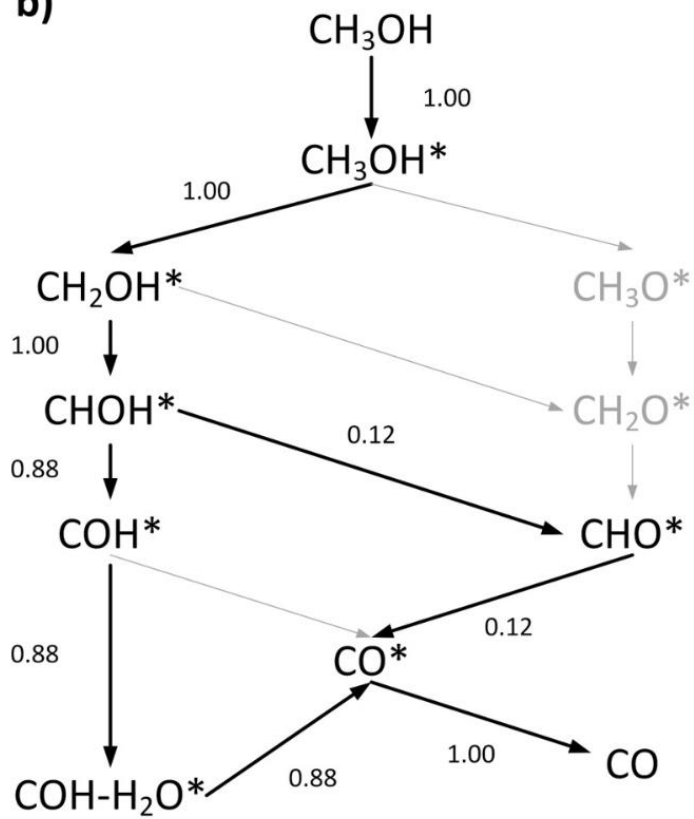
Product

Each step contributes unique information to the overall pathway.

However even branched reaction mechanisms can include only linearly independent reactions. For example often involve branched paths, for example, [García Cárcamo et al](#) found the following mechanism (and corresponding reaction path) for methanol dehydrogenation over Pt/Al₂O₃ catalysts:



b)

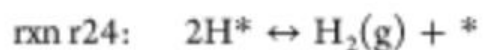
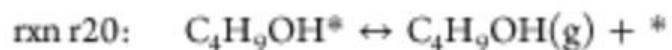
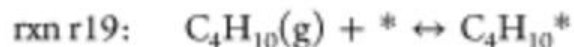
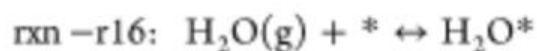
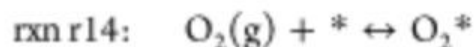
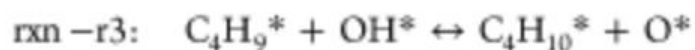


Branching Does Not Mean Dependence

A common misconception is: Branched Mechanism = Linearly Dependent Mechanism. This is not true. A mechanism may contain multiple pathways and still consist entirely of linearly independent reactions.

Key Point: Branching describes Reaction Pathways while Linear Independence describes Stoichiometric Information Content. These are different concepts.

In contrast, the mechanism of n-butane oxidation to 1-butanol by [Zhu et al](#), i.e.:



involves linearly dependent steps. It is totally to have a

mechanism where not all steps are linearly independent. In fact, this is how chemistry often operates. However, challenges arise when

Is Linear Dependence a Problem?

Not necessarily. Real chemistry often contains linearly dependent reactions. A mechanism can still be physically meaningful even when some reactions are linearly dependent.

Important Point: Linear Dependence $\neq$ Incorrect Mechanism

The issue arises when we attempt to: Analyze, Simplify, Count Degrees of Freedom, and Determine Independent Pathways because redundant reactions can obscure the underlying structure of the mechanism.

Hence, we need to be able to know how many reactions within a mechanism are linearly independent. We can do this using the procedure of

Why Do We Care About Matrix Rank?

Determining independence manually becomes difficult for large mechanisms.

Instead, we:

Construct Stoichiometry Matrix

↓

Calculate Matrix Rank

↓

Determine Number of Independent Reactions

Interpretation

Rank of Stoichiometry Matrix = Number of Linearly Independent Reactions

This provides a systematic method that works for both small and large reaction networks.

Let's do an example. This is Exercise 2.5 from Rawlings and Ekerdt: Consider a mixture of CO, H₂, and CH₄ that is fed into a furnace with O₂ and produces CO, CO₂, and H₂O. The following chemical reactions have been suggested to account for the products that form:

1. $\text{CO} + \frac{1}{2} \text{O}_2 \rightarrow \text{CO}_2$
2. $\text{H}_2 + \frac{1}{2} \text{O}_2 \rightarrow \text{H}_2\text{O}$
3. $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
4. $\text{CH}_4 + \frac{3}{2} \text{O}_2 \rightarrow \text{CO} + 2\text{H}_2\text{O}$

How many linearly independent reactions are there? Which reaction cannot be eliminated from the set? Why?

What Does It Mean for a Reaction To Be Eliminated?

Suppose one reaction can be generated by combining other reactions in the set. Then: That Reaction Adds No New Information and can be removed without changing the stoichiometric information contained in the mechanism.

Physical Interpretation

Independent Reaction



Must Be Kept

Dependent Reaction



Potentially Removable

The goal is to identify the smallest set of reactions that completely describes the chemistry.

First step:

What if you have a larger mechanism such as the one by Zhu et al? Use mathematical software to determine the matrix rank. For example, this is how it can be done in [python](#).

Why Use Software?

For a small system: 2 Reactions, 3 Reactions, 4 Reactions: linear independence can often be evaluated by hand.

For a large reaction mechanism: 10 Reactions, 20 Reactions, 100+ Reactions: manual analysis becomes impractical.

Software can rapidly calculate: Matrix Rank, Null Space, Independent Reactions, and identify redundant reaction steps.

Engineering Perspective

Reaction mechanisms often contain dozens or hundreds of elementary reactions, making computational tools essential for analysis.

Summary and Key Takeaways

Today we introduced the concept of linearly independent reactions.

Key Ideas

- ✓ A reaction is linearly independent if it cannot be reproduced using combinations of other reactions.
- ✓ Branched mechanisms can still consist entirely of linearly independent reactions.
- ✓ Real reaction mechanisms often contain linearly dependent reactions.
- ✓ Linear dependence does not mean the mechanism is incorrect.
- ✓ Matrix rank provides a systematic method for counting independent reactions.
- ✓ The rank of the stoichiometry matrix equals the number of linearly independent reactions.

✓ Software tools are often needed for large reaction networks.

Most Important Idea

Reaction Network



Stoichiometry Matrix



Matrix Rank



Independent Reactions

The central lesson of this lecture is that reaction mechanisms often contain redundancy, and matrix methods provide a powerful way to determine how many truly independent reactions are present.